

Dr Harry Patria, PhD, PMP®, PMI-CPMAI®

Visa Status: UK PhD Post-Study Work Visa (3-year, no sponsorship required) | Ready to relocate

PROFESSIONAL SUMMARY

Experienced Data and AI Lead with expertise in machine learning, predictive analytics, and AI-driven business intelligence. Over 18 years of experience transitioning from engineering to business analytics, with a strong focus on building scalable data pipelines, developing machine learning models, and deploying AI agent solutions for real-time decision-making. Proven track record in data analysis, business problem-solving, and model optimisation, driving efficiency improvements of 15-20% and enhancing predictive accuracy by 85%+. Proficient in translating complex data findings into relevant business strategy, and able to partner across functions and the organisation on AI strategy implementations.

SKILLS

Experience with statistics, programming in Python, R, SQL, big data technologies and proficient in scikit-learn, TensorFlow, PyTorch, Spark, distributed computing frameworks. Solid experience of data wrangling, feature engineering, and statistical modelling with machine learning and neural networks. Proficient in data visualisation (Power BI, Tableau, Matplotlib, Seaborn, Streamlit). Expertise in CI/CD pipelines, MLOps, cloud deployment (GCP, Azure), and model monitoring with MLflow, Docker, and Kubernetes. Skilled in A/B testing, experimental design, and optimisation.

PMP Certification	Feb 2026	View Credential
PMI Certified Professional in Managing AI (PMI-CPMAI)	Jan 2026	View Credential
Google Generative AI Leader	Jan 2026	View Credential
Microsoft Power BI Data Analyst Associate	Dec 2025	View Credential
Databricks Lakehouse Fundamentals	Feb 2026	View Credential

WORK EXPERIENCE

2026 - Present **i-Vigilant Technologies**

<https://i-vigilant.com/> | Aberdeen, Scotland, UK

Position: Principal AI Software Engineer

- Led AI modelling for the AI Smart Meter Analytics (AI-SMA) and Ultrasonic Meter (USM) platform. Framed anomaly detection as an event-stream problem across 20+ utility customers with a 94% detection rate and ~15% fewer false positives than the legacy rules engine.
- Built the analytical layer of AI-SMA covering machine learning, reasoning, summary and retrieval. Comparative evaluation across XGBoost, LightGBM and small transformer variants with 5-fold cross-validation lifted reporting accuracy ~10 percentage points over the earlier standardised baseline.
- Trained the AI Smart Reporting feature. Built the knowledge layer from instrument telemetry, academic papers and users' chain-of-thought traces, then wired LLM ontology, evaluation harnessing and graph knowledge (GraphRAG) into retrieval and summarisation to lift accuracy and cut latency.
- The reporting model now runs at Shell and BP: ~10x faster report generation and analyst effort down ~80%. Evaluated GPT, Claude, Llama and Qwen for internal energy queries at 89% eval accuracy.
- Designed the AI software architecture that won the Equinor Brazil tender (GBP 1M+), selected as sole Principal AI Software Engineer on a 4-person expert team alongside three gas metering specialists.

2021 - Present **Patria & Co. PT. Strategi Transforma Infiniti** www.patriaco.co.uk www.convergeni.com

Position: Chief Data and AI

- Led 50+ AI and data science projects, applying machine learning and predictive analytics to improve demand forecasting accuracy by up to 30%, enhance customer segmentation models by 40%, and develop recommendation systems that increased user engagement by 15%
- Built and deployed 70+ predictive models using Python, TensorFlow, PyTorch, and Scikit-learn, improving forecasting efficiency and reducing decision-making time for business leaders
- Designed and automated CI/CD pipelines, cutting model retraining time by 60% and improving AI system reliability in production environments
- Developed scalable data pipelines and feature engineering workflows, ensuring 95% data accuracy for downstream machine learning applications
- Created interactive business intelligence dashboards (Power BI, Tableau), enabling real-time data-driven decision-making that led to 20% faster response times to market changes
- Conducted A/B testing and experimental design, optimising pricing strategies and customer engagement metrics, resulting in a 12% increase in conversion rates
- Integrated cloud-based AI solutions (Google Cloud, Azure) for large-scale deployment, reducing infrastructure costs by 25% while improving model monitoring and performance metrics
- Designed corporate masterclasses on Agentic AI, Retrieval-Augmented Generation (RAG), Data Analytics, and Machine Learning, upskilling 300+ professionals in AI and data science

2022 – 2024 **Startup Campus**

www.startupcampus.id

Position: Data Science and AI Lead

- Trained 250+ professionals in machine learning, statistical modelling, and MLOps
- Guided students in building predictive models and data-driven platform (Scikit-learn and PyTorch)

- Led workshops on hypothesis testing, statistical analysis, regression, clustering, and optimisation.

2020 – 2023 PT Indeks Komoditas Indonesia a subsidiary of Coalindo Energy <https://coalindoenergy.com/>

Position: Data Scientist

- Built analytical platforms using global UN Comtrade database and extracted patterns and trendlines (CPO, minerals)
- Designed modelling and forecasting platforms to help users optimise business recommendations
- Conducted a series of workshops for analysts and stakeholders to unlock the use of analytics.

2020 BINUS University, Universitas Indonesia, Institut Teknologi Bandung, ITS Surabaya

www.binus.ac.id www.ui.ac.id www.itb.ac.id www.its.ac.id

Position: Faculty Member and Lecturer

- Performed teaching in data science, strategy, design thinking, renewable and sustainable energy, and innovation technology for 700+ graduate students at 4 leading Indonesian universities

2020 - 2021 Berdikari (Persero) – State-owned Enterprises www.ptberdikari.co.id

Position: Vice President Corporate Planning and Business Development (August – April)

- Directed strategic initiatives and corporate planning during a pandemic, driving a 15% year-on-year growth through digital transformation, diversification, and strategic partnerships
- Initiated and constructed intelligence dashboards for Sales, Operation, and Risk Management by performing clustering, customer profiling, market forecasting, and financial modeling

2018 - 2020 Elnusa Petrofin (Pertamina Group) www.petrofin.co.id

Position: Senior Manager Corporate Planning (October – March)

- Advocated for digital adoption of insights from analytical platforms among senior and mid-level management, resulting in a diversified portfolio and succeeding to 10% annual growth
- Led annual and long-term strategic planning program and budget by managing focus group discussions, management meeting reviews, scenario planning and performance management

2015 - 2018 Elektrindo Perkasa Utama and Eramas Persada Energy www.epu.co.id

Position: Senior Manager Business Development

- Formed 3 ventures through leading corporate consortia: led to auction qualification stages for 100 MW Jambi Peaker and 250 MW Riau Peaker Gas Power Plants and initiated renewable businesses of 1.5 MWp solar PV sustainable with Energy Service Company Business Model
- Enhanced investment strategy using project budgeting, financial optimisation, and what-if scenarios using Monte Carlo and Sensitivity Analyses

2010 - 2015 SKK Migas (Upstream Oil and Gas Taskforce) www.skkmigas.go.id

Pertamina Gas Negara PGN, PT Singgar Mulia Engineering

Position: Senior Analyst and Process Engineer

- Organised major LNG players (Pertamina, British Petroleum BP Tangguh, Shell, PGN, ENI) to enhance LNG export and domestic market balance
- Headed ministry-assigned teams conducting thorough technology and commercial due diligence at 5 Regasification terminals; led technical, market, and financial assessments
- Optimised Government split in 10+ Gas Sales Agreements (GSAs) via financial models and analysis for fossil fuels and renewable energy using Excel Optimiser and sensitivity analyses.

2009 - 2010 Tetra Pak Stainless Equipment www.tetrapak.com

Position: Process Design Engineer

- Designed manufacturing skids upon Engineer-To-Order by performing process CAD drawing, process calculation, process simulation, process design, Bill of Material (BoM) and project design implementation
- Coordinated with other engineering departments (electrical, instrument, mechanical) to ensure the projects were On Time, Budget, Scope, and Revenue (OTOBOSOR)

2008 - 2009 RPE-Engineering www.rpe-engineering.com

Position: **Jr. Process Engineer**

- Supported the Detailed Design (DD) and Front-End Engineering Design projects FEED (Compressor Replacement, Onshore Vessel Facilities, Offshore Gas Processing Platform) by conducting process rating, simulation, safety analysis, and process description report
- Performed process simulation with HYSYS and pipeline simulation with TGNET and Pipeline Studio to fine-tune the best configuration of Offshore-Pipeline-Onshore Facilities

EDUCATION

2022 - 2026 University of Strathclyde **Glasgow, UK**
PhD in Cognition and Scenario Planning
A Fully Funded Excellence Scholarship and Stipend, UK Research and Innovation (UKRI)
Passed the PhD viva exam on 2nd September 2026

2023 - 2025 Imperial College London **London, UK**
MSc Business Analytics, Imperial Business School

2021-2022	Newcastle University <i>MSc Data Science, School of Computing</i> <i>Grade: Distinction funded by Student Excellence Vice-Chancellor Scholarship</i> <i>Dissertation in Missingness Pattern and Exploration using Global Energy Data</i>	Newcastle, UK
2017-2020	Universitas Indonesia (1st QS Rank in Indonesia) <i>PhD Strategic Management, Faculty of Economics and Business</i> <i>A fully funded scholarship and Outstanding PhD Graduate</i> <i>Grade 3.88 out of 4.00 for 2 Years and 11 Months, Thesis in Strategic Decision in the Energy Sector</i>	
2013-2015	Universitas Indonesia <i>MSc Economics, Grade 3.79 of 4.00 Distinction</i> <i>The Best Thesis of The Year and The Best Paper in LPEM ADB Economic Forum</i> <i>Dean's List for Academic Excellence, Thesis in Oil Exploration Economics</i>	
2009-2011	Universitas Indonesia <i>MEng Chemical Engineering, Grade 3.57 of 4.00</i> <i>Dissertation in Gas Processing, Simulation, and Economic Evaluation</i>	
2003-2008	Institut Teknologi Bandung <i>BEng Chemical Engineering</i> <i>Heat propagation in reversed flow reactor: Performed experiments and process simulations.</i>	

ACHIEVEMENTS

2025	Rising Star and Alumni Award by Vice-Chancellor https://bit.ly/NewcastleAward
2022	1st Winner Data Scientist, UK Maritime Week Innovation, Newcastle, UK
2018	The Best Innovator and Employee of The Year at Elnusa Petrofin Designed and automated the BizGen platform to optimise business and operational decisions.

PUBLICATION (SELECTED)

Google Scholar: 200+ citations and an h-index of 9: <https://scholar.google.com/citations?user=OJVjCmsAAAAJ&hl=en>

- Syauqie, M., **Patria, H.**, Hastono, S. P., Siregar, K. N., & Moeloek, N. D. F. (2025). Deep learning for vision screening in resource-limited settings: development of multi-branch CNN for refractive error detection based on smartphone image. *Frontiers in Computer Science*, 7, 1576958.
- Syauqie, M., Hastono, S. P., Siregar, K. N., Moeloek, N. D. F., & **Patria, H.** (2025, May). Development of High-Accuracy Refractive Error Screening Model in Low-Resource Environments: A Hybrid Approach Using Smartphone-Based Photorefracton and Multi-Branch Convolutional Neural Network Analysis.
- Patria, H.**, & Tapinos, E. (2025). "Team Diversity in Scenario Planning: Theoretical Insights and Implications for Team Composition." In M. Crawford & G. Wright (Eds.), "Improving and Enhancing Scenario Planning" (pp. 266–285). Edward Elgar Publishing.
- Kurniawan, A., Rifa'i, A., Nafis, M. A., Andriaswuri, N. S., **Patria, H.**, & Purwitasari, D. (2022, August). Feature Selection and Sensitivity Analysis of Oversampling in Big and Highly Imbalanced Bank's Credit Data. *ICoICT* (pp. 35-40). IEEE.
- Patria, H.**, 2023. Bayesian Regression for Predicting Price Empirical Evidence in American Real Estate. *Data Science: Journal of Computing and Applied Informatics*, 7(1), pp.15-23
- Patria, H.** (2026). *Agentic AI: Building AI agents and retrieval systems, a masterclass in LLM agents, RAG, and production deployment. 1st Edition Handbook*. Convergeni. Zenodo. <https://doi.org/10.5281/zenodo.22049522>
- Patria, H.** (2026). *Agentic AI engineering handbook: A handbook for building AI software and databases*. Convergeni by Patria & Co. <https://doi.org/10.5281/zenodo.22139547>

PROFESSIONAL NETWORKS

- Project Management Institute, UK Chapter and Indonesia Chapter
- British Academy of Management (BAM) and The Operational Research Society UK
- Indonesia Strategic Management Society (ISMS) <https://isms.or.id/>
- Scenario for Tomorrow <https://scenarios.farhub.org/contributors/>